CYBERSECURITY

Old: Not designed for modern cyber threats; vulnerable to bots, scraping and hacking

New: Continuous cybersecurity monitoring across 120,000+ endpoints, including dark-web surveillance

TICKETING ABUSE

Old: Enabled large-scale ticketing black markets

New: Designed to eliminate bot manipulation and unauthorised access

USER INTERFACE

Old: No multilingual support

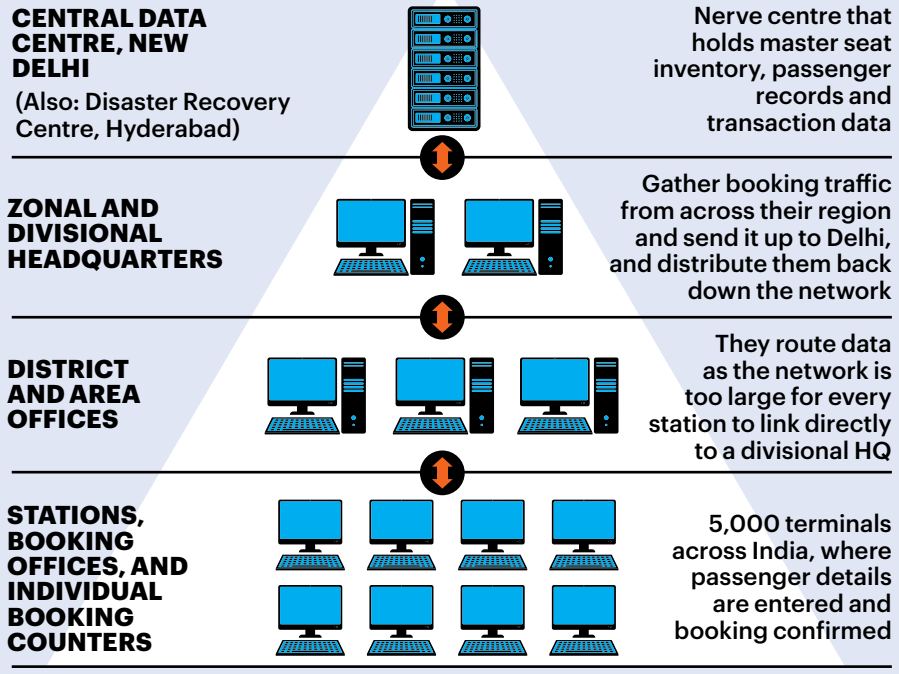
New: Multilingual interface

STRATEGIC GOAL

Old: Designed primarily to process reservation transactions

New: First step toward eventually eliminating waiting lists

HOW THE NEW TICKETING SYSTEM WORKS



A booking request travels from counter to Delhi and back in seconds. If one link fails, the system automatically reroutes

OSPF (Open Shortest Path First) routing; Ethernet WAN links, dual ISP per terminal

WHAT IT DOES

- **Instant processing of booking request**
- **Seat availability data in fast RAM**
- **Parallel data stores for speed, accuracy**
- **Fault isolation to prevent shutting the system down**
- **Fast routing across 5-tier WAN**
- **End-to-end security to curb manipulation**

Not just reserved tickets, the Unreserved Ticketing System (UTS), which handles millions of platform tickets and general-class bookings, is also being rebuilt under a Rs 200-crore project, sanctioned separately. Phase I ended in October 2021. Phase II—rebuilding the application architecture on open-standard technologies—is now underway. Full rollout is targeted for December 31.

GUARDING THE HOUSE

The effort is possible because the Railways now has both the money—given the government's sustained increase in capex every year over the past 10 years—and the political will, driven in no small measure by Union railways minister Ashwini Vaishnaw. Nowhere is this more apparent than in the second, larger component of this exercise: the Indian Railways Information Security Operations Centre (IR-SOC). A Rs 600 crore security layer, it is being designed to safeguard the rail digital infrastructure from external threats.

Globally, railway systems—from Ukraine and Belarus to Iran—have faced external threats. The IR-SOC will monitor every corner of the new digital assets for intrusions, anomalies and breaches, share threat intelligence with national agencies, such as the Indian Computer Emergency Response Team (CERT-In) and National Critical Information Infrastructure Protection Centre (NCIIPC). The SOC functions as a hub-and-spoke model: a central facility, based at Rail Bhawan in New Delhi, is at the top, feeding into 26 sub-SOCs across all zonal railway headquarters, production units and research institutions. Implementation is underway—hardware has already been installed at all but two sub-SOC locations, connectivity set up at 14 sites and 36 personnel deployed throughout the network, securing all the potentially vulnerable entry points to the network.

The scale of what the Railways hopes to address through this digital upgrade is enormous. It aims to ensure that for the next 40 years, Indian rail journeys run on a far more resilient backbone. ■